

L05 DAYLIGHT DESIGN STRATEGIES | O (MAX : 4 PT)

Intent : Provide daylight exposure indoors through design strategies.

Summary : This WELL feature requires projects to design spaces to integrate daylight into indoor environments, so that daylight may be used for visual tasks along with electric lighting. It also provides individuals with a connection to outdoor spaces through windows.

Issue : Humans spend approximately 90% of their time indoors.¹ The way buildings are designed has an enormous impact on occupant exposure to daylight.² Exposure to daylight has been proven to have a substantial impact on mood, circadian health and productivity.^{2,3} Studies have associated lack of exposure to daylight with a disruption in the circadian rhythms of humans and a decrease in quality of sleep.² Rooms with large windows also reduce recovery time for patients suffering from severe depression and those recuperating after heart attacks, compared to similarly afflicted patients in rooms with windows facing buildings or other obstructions.⁴⁻⁶ Window characteristics have also been found to have an impact on students' performance in schools.⁷ Students have been shown to perform better, when they have access to windows that can be opened.⁸ Studies also show strong links between quality of views in office environments and performance of employees.⁹ Indoor spaces with daylight exposure have been shown to have fewer bacteria, compared to spaces without any light exposure.¹⁰

Solutions : Building design and interior layout have a substantial impact on the amount of daylight in an indoor space. Indoor daylight access should be accounted for at all stages of building planning from architectural and façade design to interior design and layout. Indoor daylight planning should be coupled with glare control strategies, such as integrating shading for all windows.

PART 1 IMPLEMENT DAYLIGHT PLAN (MAX : 2 PT)

For All Spaces except Dwelling Units & Guest Rooms:

The following requirement is met:

- a. The project demonstrates that the following conditions are achieved:

Tier	Interior Layout	Façade Design	Points
1	70% of all workstations are within 7.5 m of envelope glazing. Visible light transmittance (VLT) is greater than 40%.	OR Envelope glazing is no less than 15% of the regularly occupied floor area. Visible light transmittance (VLT) of windows is greater than 40%.	1
2	70% of all workstations are within 5 m of envelope glazing. Visible light transmittance (VLT) is greater than 40%.	OR Envelope glazing is no less than 25% of the regularly occupied floor area. Visible light transmittance (VLT) of windows is greater than 40%.	2

For Dwelling Units & Guest Rooms:

The following requirement is met:

- a. One of the following requirements is met in each dwelling unit:

Tier	Vertical Envelope Glazing Requirements	Points
1	Vertical envelope glazing is no less than 15% of the regularly occupied floor area of each dwelling unit. Visible light transmittance (VLT) is greater than 40%	1
2	Vertical envelope glazing is no less than 25% of the regularly occupied floor area of each dwelling unit. Visible light transmittance (VLT) is greater than 40%	2

PART 2 INTEGRATE SOLAR SHADING (MAX : 2 PT)

For All Spaces except Dwelling Units & Guest Rooms:

The following requirements are met in regularly occupied spaces:

- a. All vertical envelope glazing has shading that meets one of the following:

Tier	Type of Shading	Points
1	Manual shading controllable by regular occupants at all times. Shades are regularly opened once a day for all days that the project is in use	1
2	Shading is automated to prevent glare	2

For Dwelling Units & Guest Rooms:

The following requirements are met in dwelling units:

- a. All vertical envelope glazing has shading that meets one of the following:

Tier	Type of Shading	Points
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1	Manual shading is controllable by regular occupants at all times.	1
2	Shading is automated to prevent glare.	2

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